

BRAINPOWER
ROMANIAN KINDER



Teacher Code :

AJIA MORK LOPF THOS

WONJ TRIA CHAS HEWP



MATHEMATICAL METHODS UNIT 2

Tech-Free

Test theme: THE BACKROOMS

Reading Time: 5 minutes

Writing time: 45 minutes

QUESTION AND ANSWER BOOK

<i>Number of questions</i>	<i>Number of questions to be answered</i>	<i>Number of marks</i>
6	6	31

● Students are permitted to bring into the test room: pens, pencils, highlighters, erasers, sharpeners, rulers.

● Students are NOT permitted to bring into the examination: notes of any kind, blank sheets of paper, white out liquid/tape or a calculator of any type.

Materials supplied

- Question and answer book.
- Working space is provided throughout the book.

Instructions

- Write your **name** in the space provided above on this page.
- All written responses must be in English or Romanian.

Students are NOT permitted to bring mobile phones and/or any other unauthorised electronic devices into the test room.

Instructions

Answer **all** questions in the spaces provided.

Unless otherwise specified an **exact** answer is required to a question.

In questions where more than one mark is available, appropriate working **must** be shown.

**Question 1** (5 marks)

*You have been walking for what feels like 14 hours through the endless, non-Euclidean maze of yellow-wallpapered offices. You are a **Lost Wanderer**, a victim of a reality glitch. The monotypical wallpaper is peeling in a way that suggests a spatial fold. The fluorescent lights hum at a frequency of **60 Hz**, creating a sensory prison. To 'noclip' through the corner of a Manila Room and find a temporary refuge, you must calculate the exact orientation of the hallway.*

- a. Calculate the exact value of the following spatial coordinate:

$$\sin\left(-\frac{17\pi}{6}\right) \cdot \cos\left(\frac{13\pi}{4}\right)$$

2 marks

- b. Evaluate the following expression to stabilise your pulse against the deafening hum. Ensure the denominator is rationalised:

$$\tan^2\left(\frac{2\pi}{3}\right) - \frac{1}{\cos^2\left(\frac{11\pi}{6}\right)}$$

3 Marks

**Question 2** (4 Marks)

*You successfully noclip through the floor and tumble into a massive, dingy warehouse. This is Level 1. You find a crate of Almond Water; but a **Hound**, a mangy, humanoid creature on all fours, is guarding it. You notice a damp stain on the concrete floor that forms a right-angled triangle relative to the iron support pillars.*

Let θ be the angle of the stain.

Given that $\cos(\theta) = -\frac{4}{5}$ and $\pi < \theta < \frac{3\pi}{2}$:

- a. To move silently, you must calculate the vertical component of your step. Determine the exact value of $\sin(\theta)$

2 Marks

- b. The Hound turns its head. You must calculate the supplementary angle to remain in its blind spot. Find the exact value of $\tan(\pi - \theta)$

2 Marks



Question 3 (6 Marks)

*You escape the Hound and enter Level2: a suffocatingly narrow maintenance tunnel. The air is boiling hot, and the hiss of high-pressure steam is deafening. Two steam pipes, Pipe A and Pipe B, are vibrating. Their displacement follows the curves of the base reality. A **Smiler**, a glowing, razor-sharp grin in the dark, is floating toward you. It only strikes when the pipe displacements are equal*

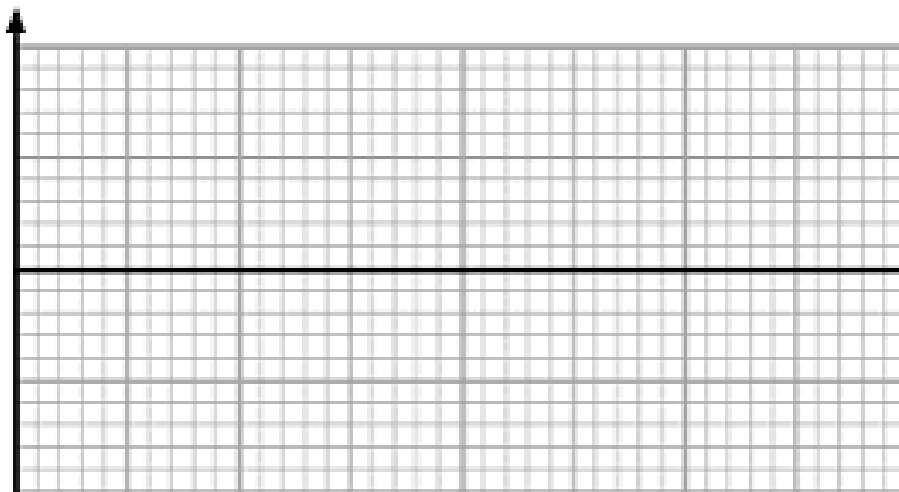
Let Pipe A be modelled by $f(t) = \sin(t)$ and Pipe B by $g(t) = \cos(t)$

- a. Find the exact times $t \in [0, 2\pi]$ when the Smiler will strike

2 Marks

- b. You need to map the safe zones. On the same axes, sketch $y = f(t)$ and $y = g(t)$ for $t \in [0, 2\pi]$. Label all axial intercepts and the points of intersection with their exact coordinates

4 Marks



**Question 4** (4 Marks)

*The steam pipes burst, and you are sucked into a drainage grate, waking up in the serene, white-tiled chambers of the Poolrooms. The crystal-clear water ripples as a **Skin-Stealer** mimics the sound of a human survivor from behind a pillar. The Skin-Stealer's movement follows a transformed path:*

$$h(x) = -\sqrt{2}\cos\left(2x + \frac{2\pi}{3}\right) + \sqrt{6}$$

- a. State the period and amplitude of the Skin-Stealer's movement $h(x)$

2 Marks

- b. To predict its next lunge, state the sequence of transformations required to map the graph of $y = \cos(x)$ to the graph of $y = h(x)$

2 Marks

**Question 5** (6 Marks)

The clean tiles of the Poolrooms begin to de-rez, turning back into a dark, corrupted version of Level 0. A 'Bacteria' entity is mimicking a distorted shriek. The geometry is failing. To stabilise your physical form and prevent being 'deleted', you must solve the frequency of the flickering of the flickering lights

- a. Find the general solution for x in the following equation:

$$2\sin\left(3x - \frac{\pi}{4}\right) + \sqrt{2} = 0$$

4 Marks

- b. The exit is near. Find all specific solutions for x in the interval $x \in [-\pi, 0]$ to fix the coordinate

2 Marks



Question 6 (6 Marks)

You reach a massive, glowing arcade, the final hub before you exit. The 'Log Out' button the terminal is glitched. To trigger the exit protocol, you must solve the following that governs the de-rezzing of the Backrooms' walls.

- a. Solve the following equation $x \in [0, 2\pi]$ to unlock the terminal:

$$2\sin^2(x) + 3\cos(x) - 3 = 0$$

4 Marks

- b. The exit portal will only stabilise if $2\sin^2(x) + 3\cos(x) - 3 \leq 0$. Using your results from part (a), state the intervals $x \in [0, 2\pi]$ for which you safely noclip back into reality

2 Marks

**THE WALLS ARE DE-REZZING. THE HUMMING HAS STOPPED. THAT IS A
BAD SIGN.**

END OF TEST... UNTIL THE NEXT NOCLIP...